

ORIGINAL RESEARCH

Bias due to interval censored outcomes in a study of flare risk after hydroxychloroquine taper/cessation in systemic lupus erythematosus

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Abstract

Objective: This study attempted to quantify the bias expected due to partly interval-censored (IC) outcomes in the estimated association between hydroxychloroquine (HCQ) taper/cessation and time to disease flare among individuals with systemic lupus erythematosus (SLE).

Methods: Using data-driven simulations, we estimated bias expected due to IC using real-world data from the Systemic Lupus International Collaborating Clinics inception cohort. The time-varying exposure of interest was a binary indicator of HCQ tapering/cessation. The composite outcome was lupus flare, defined as lupus hospitalizations or increases in disease activity or medication dose. The two latter components were IC, as they were recorded only at annual assessment, without a precise date. For the unknown IC event times, a “true” event time was randomly generated from a uniform distribution of the time between two assessments. Each simulated sample was analyzed separately imputing unknown event times (for IC outcomes) either at the midpoint or endpoint of the interval between the two adjacent yearly assessments. Results of multivariable Cox proportional hazards models, adjusted for demographics, drugs, and clinical variables, using either “true” or imputed IC event times were compared.

Results: The 1543 SLE patients were followed for a median of 42.2 months. During follow-up, 396 participants tapered/stopped HCQ and 1187 experienced a flare. The adjusted uncorrected hazard ratio was 1.51 (95% confidence interval: 1.30, 1.75) and 1.40 (95% confidence interval: 1.21, 1.62) for midpoint and endpoint imputations, respectively. Data-driven simulations showed that imputation of IC event times resulted in a small but systematic bias toward the null that was consistently larger for endpoint than for midpoint imputation.

Conclusions: IC events induced bias toward the null in the estimated association between HCQ taper/cessation and lupus flares. Data-driven simulations are useful for quantitative bias analyses in complex situations, as they allow accounting for relevant characteristics of a particular real-world dataset. © 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords: Pharmacoepidemiologic methods; Quantitative bias analysis; Data-driven simulations; Interval censoring; Systemic lupus erythematosus; Hydroxychloroquine

1. Introduction

Systemic lupus erythematosus (SLE) is a multisystem autoimmune disease [1,2]. Hydroxychloroquine (HCQ) is a drug recommended for all SLE patients unless contraindicated [1,3]. Despite HCQ's effectiveness at reducing flares and improving outcomes [3–6], 20%–30% of patients discontinue the drug [7]. Common reasons for discontinuation are toxicity (eg, retinopathy), patients' concerns about side effects, and disease quiescence [8–10]. Yet, HCQ treatment discontinuation or dose reduction may increase risk of SLE flares [8, 9].

Previous studies of HCQ taper/cessation used time-to-event analyses [11]. Yet, many SLE flares are identified only retroactively at clinic visits, with exact timing of their occurrence being unknown. Such events, which truly could have occurred at any time during the interval between consecutive study visits, are interval-censored (IC) [10]. During analyses, these unknown event times are commonly imputed at either the endpoint or midpoint of the

corresponding time interval. However, this can lead to inaccurate estimates [10], especially for time-varying exposures (TVEs) [11]. The resulting bias depends on several features of the empirical dataset, including (i) (most importantly) frequency of visits, (ii) definition and distribution of the TVE, (iii) strength of the exposure-outcome association, and (iv) distribution of event times [12]. As outlined in the discussion, specialized methods to handle IC event times do exist, but are difficult to apply to our specific setting, which involves both TVE and the fact that only about 50% of events are IC.

Our study assesses the expected bias due to IC outcomes, in the estimated association between the TVE representing HCQ tapering/cessation and time to SLE flare, using quantitative bias analysis [13] based on data-driven simulations [12]. This approach allows objectively quantifying the expected bias, induced by a specific data imperfection, in the results of a particular real-world study, and handles TVEs in time-to-event analyses [12].

What is new?**Key findings**

- Data-driven simulations allowed us to quantify bias due to interval censoring of event times (a common limitation of outcomes data).
- Our analyses showed that interval censoring resulted in bias that attenuated hazard ratio estimates; this was less pronounced when interval censored event times were imputed at the midpoint between two assessments rather than the later of the two assessments.

What this adds to what is known?

- Our study highlights the need to consider interval censoring of outcome events, and that quantitative bias analyses are needed to assess the impact of interval censoring.

What is the implication and what should change now?

- Data-driven simulations should be considered when dealing with data imperfections, like interval censoring of events, as simulations can be used to reflect specific features of a real-world dataset.

2. Methods**2.1. Data source**

The data stem from the Systemic Lupus International Collaborating Clinics (SLICC) inception cohort [14,15] of patients, aged 18+, who meet the American College of Rheumatology (ACR) revised classification for SLE [16]. Participants were enrolled in 1999–2012, within 15 months of SLE diagnosis, and assessed at baseline and annual visits with their treating physicians, who belong to the SLICC network [14,15]. Participants were followed until death, end of study (January 2019), or loss to follow-up. From the 1848 participants, we removed 183 who never received HCQ, and 122 with only baseline data, leaving 1543 persons for the analyses.

2.2. Exposure

The binary (yes/no) TVE is HCQ tapering or cessation. The TVE value switches from initial 0 (HCQ maintenance) to 1 at the first month when the recorded average daily HCQ dose is lower than the initial dose. After the initial tapering/cessation, the TVE cannot switch back to 0, and any dose augmentation or restart of HCQ treatment is

considered a flare (see Outcomes section). Figure 1 illustrates how the TVE is defined for participants with different patterns of changes in HCQ treatment.

2.3. Outcome

The time-to-event composite outcome, SLE flare, was defined as time to the first occurrence of: (i) SLE-related hospitalization, identified using hospitalization reasons in the SLICC database, (ii) therapy augmentation, or (iii) increased disease activity [17,18]. Therapy augmentation was defined as a new start or increased dose for any of the following lupus medications: HCQ (or chloroquine), prednisone, immunosuppressive agents (azathioprine, methotrexate, mycophenolate, cyclophosphamide, etc.), or biologics (rituximab, belimumab, etc.) [17,18]. Increased disease activity was defined as an increase of at least four points in the Systemic Lupus Erythematosus Disease Activity Index-2000 (SLEDAI-2K) score from the previous annual visit [19,20].

The database includes exact dates for hospitalizations and new therapy prescriptions. However, the exact dates for flares corresponding to increases in an existing therapy or in SLEDAI-2K score are unknown. These flares are interval censored because they are recorded only at a specific clinic visit, though, in reality, they occurred sometime in the interval between this visit and the prior visit.

2.4. Covariates

Clinically relevant covariates were selected a priori based on the literature. Sociodemographic variables included sex, age at SLE diagnosis (dichotomized at 45 years, to approximately correspond with menopause, which impacts SLE outcomes [21, 22]), race (non-Caucasian vs Caucasian), and region (North America, Europe, or Asia). We also included enrollment year (before 2006 versus 2006+), time between diagnosis and cohort entry, time-varying end-stage renal disease (based on the SLICC/ACR damage index [23]), and indicators of ever smoker and baseline use of prednisone, immunosuppressors and biologics, which may be associated with both HCQ dose and risk of flare.

2.5. Statistical analyses

In the time-to-event analyses, participants entered the risk set at the later of either their first SLICC cohort visit or first HCQ prescription to avoid immortal time bias [24]. To account for HCQ use before cohort entry, the delayed entry approach was used [25]. For example, follow-up time of a patient who started HCQ 4 months before cohort entry begins at 4 months. Accordingly, the time axis is defined as time (in months) since the first HCQ prescription (see Fig 1). Thus, across follow-up times t , the hazards of a flare are compared for patients who all started HCQ t months ago [25].

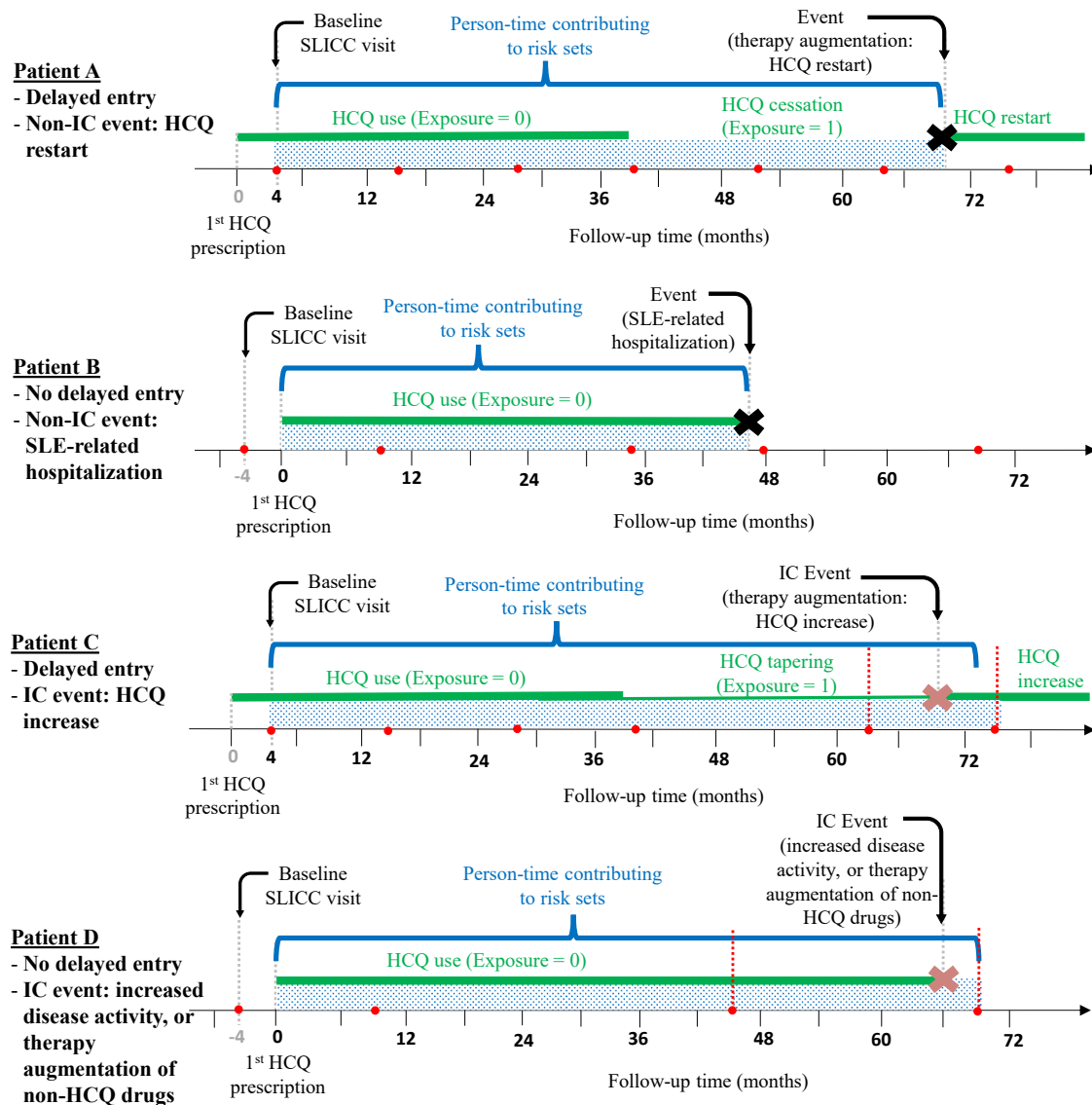


Figure 1. Illustration of follow-up and exposure definition for four hypothetical patients. Patients A and C started HCQ (t_0) 4 months before the baseline SLICC visit (cohort entry), and therefore had a delayed entry and contributed to risk sets starting from month 4. Patients B and D, however, started HCQ 4 months after baseline SLICC visit and then did not have delayed entry. Non—interval-censored (non-IC) events are illustrated by patients A and B, who respectively had an HCQ restart following an HCQ cessation and an SLE-related hospitalization. Another type of non-IC event would be a new start of any other lupus medication (not shown here). Patients C and D both show types of IC events: patient C illustrates an HCQ increase (therapy augmentation) after an HCQ tapering; and patient D shows either increase in disease activity or therapy augmentation involving lupus medications other than HCQ. HCQ, hydroxychloroquine; SLICC, Systemic Lupus International Collaborating Clinics; SLE, systemic lupus erythematosus. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

The association between the TVE representing HCQ taper/cessation and time to SLE flare was estimated using a multivariable Cox proportional hazards (PHs) model, adjusting for the above covariates. In alternative models, the unknown times of IC flares were imputed at either the endpoint or midpoint of the follow-up year during which the event was reported [26].

2.6. Design and methods of data-driven simulations

We relied on data-driven simulations [12], designed to reflect the characteristics of our SLICC cohort, to objectively

quantify the expected impact of imprecise timing of IC flares on their estimated association with time-varying HCQ taper/cessation. Our simulations followed the seven-step approach proposed by Abrahamowicz et al [12].

Step 1 (identifying the data imperfection): In the dataset, exact dates are unknown for 46% of flares, defined as increases in either disease activity or therapy. These IC flares have truly occurred at some time between the two consecutive annual visits.

Step 2 (initial uncorrected analyses): We ran two multivariable Cox PH models to get the initial estimates of adjusted hazard ratios (HRs) for exposure and all

covariates. The unknown times of IC flares were imputed at either the endpoint or the midpoint of the interval between the two relevant visits.

Step 3 (assumptions for data generation): Clinical considerations, our initial analyses and previous findings [17,18], all suggest that HCQ taper/cessation is associated with moderately increased flare risks. Thus, in four alternative scenarios, we assumed either small to moderate hazard increases ($HR = 1.2, 1.5$, or 2.0) or no association ($HR = 1.0$), which was added to assess whether imputation of IC event times might have induced a spurious association [12]. For covariates, we assumed HRs equal to adjusted estimates from the initial multivariable model (step 2) [12]. Finally, in main simulations, based on clinical considerations, we assumed that the IC flare events were uniformly distributed throughout the relevant time interval.

Step 4 (generation of “oracle” datasets, free of the imperfection): To allow comparisons with imperfect datasets (step 5), “oracle” datasets, free of the imperfection identified in step 1 [12], were generated using the following four substeps. The first two substeps used the same data for all 1000 simulation repetitions, but substeps 4.3 and 4.4 were repeated independently for each repetition.

4.1: The observed values of the TVE and all covariates were directly copied to each simulated dataset, preserving their multivariate distributions and associations with each other.

4.2: For non-IC flares, the reported event times were also copied from the real-world dataset.

4.3: For IC flares, the “true” event time was randomly generated from a uniform distribution over the between-visits interval in which the flare was originally recorded (see step 3). In sensitivity analyses, “true” times of IC flares were generated from three nonuniform beta distributions, with events more frequent in: (i) early phase, (ii) late phase, or (iii) both tails of the corresponding interval (Supplement S.1 provides more details).

4.4: Using the permutational algorithm (Supplement S.2), validated for simulating event times conditional on TVEs [27,28], we reassigned all events to subjects using weighted sampling with probabilities proportional to their individual true HRs for a given simulated scenario (with the exposure HR corresponding to one of the four values specified in step 3) [12].

Step 5 (generation of imperfect datasets): Each oracle dataset generated in step 4 was modified assuming exact times of IC events are unknown, and were, therefore, imputed, using two alternative approaches: at either the endpoint or midpoint of the 12-month interval in which the flare must have occurred. This interval was chosen because SLICC cohort visits were prescheduled every 12 months [14,15]. We assumed that patients having an IC flare would not miss their next visit. The TVE and covariate values were unchanged from the corresponding oracle dataset.

Step 6 (analyses of simulated datasets): All oracle and imperfect datasets were analyzed separately using a multivariable Cox PH model with the TVE and all covariates, and, respectively, either “true” times of IC flares or times imputed in step 5. This resulted in 3 HR estimates for the TVE for each simulation repetition.

Step 7 (summarizing the results): The true $\log(HR)$ (β) for HCQ taper/cessation, used to simulate the data, was compared to the $\log(HR)$ estimates for each of the 3 models outlined in step 6, across 1000 simulation repetitions for each scenario. Summary statistics included the absolute bias ($\text{mean}(\hat{\beta}) - \beta$) and relative bias ($[\text{mean}(\hat{\beta}) - \beta]/\beta$), SD of 1000 estimates, the root mean squared error ($\text{RMSE} = \sqrt{\text{bias}(\hat{\beta})^2 + \text{var}(\hat{\beta})}$), and the coverage rates (the proportion of 95% confidence intervals (CIs) which contained the true β). In addition, we compared the corresponding $\hat{\beta}$ estimates from the three models with alternative IC event times.

All simulations and analyses were conducted using R (version 4.4.1).

R code implementing our data-driven simulations and a generated dataset with a structure similar to our SLICC dataset are available as [Supplementary Materials](#).

3. Results

Among the 1543 participants, followed an average of 42.2 months, 1187 (76.9%) experienced a flare, including 545 (46%) IC flares, and 396 (25.7%) tapered or stopped HCQ. Table 1 summarizes cohort characteristics. Most participants were female (89.2%) and North American (63.3%). They were evenly split between Caucasian and non-Caucasian; and 20% were diagnosed with lupus beyond 45 years of age.

Table 2 summarizes results of the two initial Cox PH models, with alternative simple imputation strategies for the times of IC flares. The uncorrected adjusted HRs for the association between HCQ taper/cessation and SLE flares are 1.40 (95% CI: 1.21, 1.62) and 1.51 (95% CI: 1.30, 1.75) for the endpoint and midpoint imputations, respectively. Non-Caucasians and participants on prednisone and immunosuppressors at baseline had increased hazards of flare, whereas Asian region, compared to North America, and older age at diagnosis were associated with a lower hazard (Table 2).

Table 3 summarizes the results of main data-driven simulations with uniformly distributed generated “true” times of IC events. As expected, for the oracle datasets, the mean adjusted exposure $\log(HR)$ estimates are always very close to the corresponding true $\log(HR)$. In contrast, except for the null scenario with true $\log(HR) = 0$, the mean estimates based on imperfect datasets are slightly but consistently below the true $\log(HR)$ values, which are excluded from the 95% CIs for the mean estimates

Table 1. Descriptive statistics for the original Systemic Lupus International Collaborating Clinics (SLICC) dataset ($N = 1543$)

Variable	N (%)
Sex	
Female	1377 (89.2)
Age at diagnosis	
Over 45 years	312 (20.2)
Race/ethnicity	
Non-Caucasian	763 (49.5)
Region	
North America	976 (63.3)
Europe	418 (27.1)
Asia	149 (9.7)
HCQ taper/cessation	396 (25.7)
Enrollment year	
2006 or later	768 (49.8)
Time between diagnosis and enrollment (months)	
Mean $\pm$ SD	5.7 $\pm$ 4.3
Ever smoker	593 (38.4)
Prednisone at baseline	1000 (64.8)
Immunosuppressors at baseline	643 (41.7)
Biologics at baseline	15 (1.0)
End-stage renal disease	10 (0.7)

HCQ, hydroxychloroquine; BMI, body mass index.

(Table 3). The estimates based on endpoint imputation are systematically somewhat more biased toward the null than the corresponding midpoint estimates. Because SDs for all three estimation models are almost identical, higher bias of endpoint estimates results in slightly lower overall accuracy (higher RMSE), and lower coverage rates of the 95% CI (Table 3).

Table 4 compares the accuracy of the midpoint- versus endpoint-based estimates across alternative distributions of generated “true” IC event times, including three nonuniform beta distributions (see Supplement S.1 for details). Similar to Table 3, midpoint imputation of unknown times of IC events usually leads to improvements in the estimates’ accuracy compared to the conventional endpoint estimation which, for most simulated scenarios, result in slightly higher coverage of the 95% CIs and smaller RMSEs (Table 4). Interestingly, the general pattern of results is remarkably stable, regardless of what distribution of true IC event times is assumed. Still, as expected, the bias of the endpoint estimates is slightly reduced if the events are more likely to occur later in the between-visits interval (Table 4). Furthermore, the relative advantages of midpoint imputation are more pronounced for scenarios with stronger TVE effects, where imputation results in higher biases toward the null. Indeed, for true HR = 2.0,

in more than 80% of simulation repetitions, the oracle estimate, based on true event times, is closer to the midpoint than to the endpoint estimate, regardless of the assumed distribution (last column of Table 4).

Finally, for all scenarios with true HR = 1.0, for both midpoint and endpoint estimates (i) the 95% CIs included true value in at least 93.7% of simulated samples (eg, see coverage rates in Table 3) and (ii) in no samples the point estimates were as high as 1.40 or 1.51 (from the uncorrected analyses) reported in Table 2 (data not shown). Thus, the initial uncorrected estimates are very unlikely to represent a type I error.

4. Discussion

We sought to objectively quantify the bias expected due to interval censoring of about 50% of lupus flares in their association with a TVE, HCQ taper/cessation. Data-driven simulations, which accounted for salient features of our dataset [12], indicated that, as expected, imputation of unknown times of IC flares resulted in a systematic (albeit small) underestimation of the association’s strength. Yet, regardless of the assumptions about the TVE’s true HR and the distribution of IC event times, the midpoint imputation model yielded consistently lower bias than the endpoint imputation, resulting usually in a better bias-variance trade-off (lower RMSE) and higher coverage rates. Importantly, these simulation results help validate the conclusion that HCQ taper/discontinuation is associated with about 50%, or slightly more, higher hazard of SLE flares, as suggested by the initial midpoint imputation model. Additional corroboration of a systematic association comes from the fact that all “null” simulated scenarios, with true HR = 1.0, consistently indicated a very low risk of type I error.

Imprecise timing of events often arises in epidemiological studies [12,29], but is rarely addressed in real-world time-to-event analyses [30]. Indeed, studies of lupus flares likely have frequent IC events, but this is seldom acknowledged. Our results underscore the importance of trying to quantify the expected impact of imprecise timing of IC events, especially when studying TVEs [26,31,32]. The general pattern of our simulation results is consistent with previous findings indicating that IC events induce a systematic underestimation of the association’s strength, with stronger bias induced by the endpoint vs. the midpoint imputation of unknown event times [12,26]. However, the numerical results of our data-driven simulations reflect particular features of the SLICC dataset. Specifically, the relatively modest biases seen with either imputation strategy may be largely due to a combination of the following: (i) the fact that the proportion of IC events was only 46%; (ii) the relatively short intervals between annual assessments; and (iii) TVE definition that allowed the individual

Table 2. Initial adjusted hazard ratios based on multivariable Cox models with events imputed either at the endpoints or the midpoints of the relevant intervals

Variable	Midpoint imputation		Endpoint imputation	
	Adjusted HR	95% CI	Adjusted HR	95% CI
HCQ taper/cessation	1.51	(1.30, 1.75)	1.40	(1.21, 1.62)
Ever smoker	1.08	(0.95, 1.22)	1.08	(0.95, 1.22)
End stage renal disease	0.93	(0.38, 2.24)	0.97	(0.40, 2.35)
Sex				
Female	1.11	(0.92, 1.34)	1.11	(0.92, 1.34)
Race				
Non-Caucasian	1.26	(1.11, 1.44)	1.28	(1.13, 1.46)
Prednisone at baseline	1.17	(1.02, 1.34)	1.17	(1.02, 1.34)
Immunosuppressors at baseline	1.19	(1.05, 1.35)	1.19	(1.05, 1.35)
Biologics at baseline	0.87	(0.47, 1.63)	0.90	(0.48, 1.68)
Time from diagnosis to cohort entry (continuous)	1.00	(0.99, 1.02)	1.00	(0.99, 1.02)
Age at diagnosis				
Over 45 years	0.83	(0.72, 0.97)	0.83	(0.72, 0.97)
Region				
North America	Reference			
Europe	0.93	(0.82, 1.07)	0.93	(0.81, 1.07)
Asia	0.56	(0.45, 0.69)	0.54	(0.44, 0.67)
Enroll year				
2006 or later	0.91	(0.81, 1.02)	0.90	(0.80, 1.01)

CI, confidence interval; HCQ, hydroxychloroquine; HR, hazard ratio.

Reference categories: Sex, male; Race, Caucasian; age at diagnosis, 45 years or less; enroll year, 2005 or earlier.

exposure to change only once during follow-up. Jointly, features (ii) and (iii) implied that, for most participants, the updated exposure value in a given between-visits interval was not affected by the exact time at which the relevant IC event was imputed. Indeed, feature (iii) may explain why we found relative biases below 10%, in contrast to previous data-driven simulations that assessed a frequently changing TVE defined as any use of a drug

in the past 2 weeks, where midpoint and endpoint imputations resulted in relative biases of at least 25% and 34%, respectively [12].

Bias is a constant concern in epidemiological research, but applications of quantitative bias analysis (QBA) are still uncommon, although increasing with time [30,33]. Bias related to interval censoring of events (that can only be established retrospectively, at the time of clinic visits) is

Table 3. Results of data-driven simulations assuming uniform distribution when generating “true” interval censored event times

True HR [log(HR)]	Event times	Mean log(HR) estimate	95% CI for mean estimate	Bias	Relative bias, %	SD	RMSE	Coverage rate of
								95% CI, %
1.0 [0.000]	Oracle	0.002	(−0.003, 0.007)	0.002	–	0.078	0.078	94.7
	Midpoint	0.000	(−0.005, 0.005)	0.000	–	0.078	0.078	94.4
	Endpoint	0.007	(0.003, 0.012)	0.007	–	0.076	0.077	94.3
1.2 [0.182]	Oracle	0.180	(0.176, 0.185)	−0.002	−1.1	0.076	0.076	95.9
	Midpoint	0.173	(0.169, 0.178)	−0.009	−5.0	0.077	0.077	95.1
	Endpoint	0.172	(0.168, 0.177)	−0.010	−5.4	0.076	0.076	94.8
1.5 [0.405]	Oracle	0.410	(0.406, 0.415)	0.005	1.1	0.075	0.075	95.0
	Midpoint	0.395	(0.390, 0.399)	−0.011	−2.6	0.074	0.075	95.5
	Endpoint	0.385	(0.381, 0.390)	−0.020	−5.0	0.073	0.075	94.9
2.0 [0.693]	Oracle	0.696	(0.692, 0.701)	0.003	0.5	0.073	0.073	95.2
	Midpoint	0.672	(0.667, 0.676)	−0.021	−3.1	0.072	0.075	94.3
	Endpoint	0.651	(0.646, 0.655)	−0.042	−6.1	0.070	0.082	91.6

CI, confidence interval; IC, interval censored; HR, hazard ratio; RMSE, root mean square error; SD, SD of 1000 log(HR) estimates.

Table 4. Comparison of midpoint *versus* endpoint-based estimates across selected simulated scenarios

Distribution of generated IC event times	True HR [log(HR)]	Relative bias, % (95% CI); midpoint	Relative bias, % (95% CI); endpoint	Coverage rate of 95% CI, %; midpoint	Coverage rate of 95% CI, %; endpoint	RMSE ratio midpoint/endpoint	% Oracle dataset closer to midpoint than to endpoint ^a
Uniform	1.2 [0.182]	−5.0 (−7.6, −2.4)	−5.4 (−8.0, −2.8)	95.1	94.8	1.006	56.7
	1.5 [0.405]	−2.6 (−3.8, −1.5)	−5.0 (−6.1, −3.9)	95.5	94.9	0.996	63.8
	2.0 [0.693]	−3.1 (−3.7, −2.4)	−6.1 (−6.7, −5.5)	94.3	91.6	0.914	83.3
Early skewed	1.2 [0.182]	−4.7 (−7.4, −2.1)	−4.2 (−6.8, −1.6)	95.3	95.0	1.029	60.6
	1.5 [0.405]	−3.0 (−4.1, −1.9)	−5.7 (−6.8, −4.6)	95.6	94.7	0.979	67.2
	2.0 [0.693]	−3.6 (−4.2, −2.9)	−6.9 (−7.6, −6.3)	94.5	91.0	0.903	83.1
Late-skewed	1.2 [0.182]	−3.1 (−5.8, −0.5)	−3.7 (−6.2, −1.2)	95.0	95.2	1.035	53.3
	1.5 [0.405]	−2.4 (−3.5, −1.3)	−4.5 (−5.5, −3.4)	96.0	95.4	1.017	62.0
	2.0 [0.693]	−2.6 (−3.3, −2.0)	−5.4 (−6.1, −4.8)	94.2	91.9	0.948	80.3
U-shape	1.2 [0.182]	−1.8 (−4.5, 0.8)	−2.4 (−5.0, 0.3)	94.8	94.7	1.012	56.0
	1.5 [0.405]	−4.0 (−5.1, −2.8)	−6.6 (−7.7, −5.5)	94.6	94.6	0.990	64.3
	2.0 [0.693]	−3.5 (−4.1, −2.8)	−6.5 (−7.1, −5.9)	94.2	90.8	0.911	80.7

CI, confidence interval; IC, interval-censored; HR, hazard ratio; RMSE, root mean square error.

^a Calculated as the percentage of 1000 simulation repetitions for which the absolute value of the difference between the midpoint and oracle estimates is smaller than the corresponding difference between the endpoint and oracle estimates, that is, $|l_{\text{midpoint}} - o_{\text{oracle}}| < |l_{\text{endpoint}} - o_{\text{oracle}}|$.

relevant for many diseases. Data-driven simulations allow an objective quantification of the expected impact of interval censoring on the results of a particular study, and can be employed for multivariable survival analyses with TVE, where existing QBA approaches may be difficult to implement [34,35]. Still, it should be emphasized that our data-driven simulations can only assess the expected bias but not correct for it.

Therefore, future research should attempt to adapt to our specific setting some of the elegant methods proposed for handling IC events in the statistical literature. However, the relevant method has to accommodate two specific features of our data: (i) TVE and (ii) only about 50% of the events are IC. In fact, excellent reviews of methods for IC events do not discuss TVEs [10,29], which are not handled by either the seminal Turnbull's nonparametric MLE method [36] or the estimating equations-based approach [37]. Similarly, papers that proposed multiple imputation of the missing times for IC events do not address either TVEs [38,39] or partly IC events [40]. Importantly, Zeng et al's [41] highly cited extension of the classic Finkelstein's approach [42] was recently adapted to partly IC events [43]. However, both methods [41,43] and another approach that also handles partly IC events [32], all assume TVEs may only change their

values at visit times, which does not hold for pharmacoepidemiological exposures. Finally, popular parametric models for IC events with TVEs do not handle partly IC events [31]. The same issue is relevant for recent papers adapting the simulation-extrapolation methodology to correct for biases induced by unknown timing of IC events [26,44].

A potential limitation of our analyses concerns exposure definition, which combines HCQ tapering with discontinuation, that was assessed separately in a previous study [17]. Our definition was based on clinical considerations and empirical results. Clinically, both HCQ tapering and cessation may represent patient nonadherence, and may be difficult to distinguish, especially for patients who started on a very low dose. Furthermore, in preliminary analyses the separate adjusted HR estimates of 1.38 (95% CI: 1.16, 1.64) for tapering and 1.42 (1.17, 1.74) for cessation were very close.

5. Conclusion

Our work should encourage other researchers to implement data-driven simulations and other relevant QBA tools in their real-world studies, to quantify the expected strength of the bias due to interval censored events.

CRediT authorship contribution statement

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Declaration of competing interest

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Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112261>.

Data availability

The authors do not have permission to share data.

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